

Approximating Irrational Numbers

Teacher's Edition

Good News About Lesson 2

Math Nation's own Lesson 4 comes down on our side of the square-root question.

In their 2.4.2, Damion writes four equations and students judge them. Equation A is $\sqrt{64} = 8$ — one value, the positive one — and their key points it **correct**. That is the opposite of their Lesson 2, where $\sqrt{64} = \pm 8$ appears throughout.

Their teacher note even anticipates the collision: “some students may also identify Equation A as an incorrect answer because only one of the two solutions is listed.” A student who learned their Lesson 2 properly is now expected to be talked out of it.

Then their 2.4.4 introduces the term formally — **principal square root** — as a Key Term for this lesson.

What this means for your class. Your students learned the correct definition four days ago. They will agree with 2.4.2 without a discussion, and the naming in 2.4.4 lands as “here is the word for the thing you already know” rather than as a correction. Nothing in this lesson needed changing on that point.

One small repair was made. Their definition reads “the positive square root of a positive real number,” which leaves out zero — $\sqrt{0} = 0$, and 0 is not positive. The word is **non-negative**, and the Lesson 2 worksheet already had $\sqrt{0} = 0$ on it.

Standard and Targets

MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.

Boundary Approximate numerical expressions involving one arithmetic operation, estimating square roots or π . Every expression on these sheets is one operation.

Also touched MA.8.NSO.1.2, plotting and comparing — which is Lesson 5's benchmark and arrives properly tomorrow.

Learning targets I can tell a rational number from an irrational one. I can approximate a square root or cube root to the nearest tenth. I can plot the result on a number line.

Essential questions Which roots are irrational, and how do you know? How do you get a decimal for a number whose decimal never ends?

Building on Lessons 1 to 3 for the two lists of perfect squares and perfect cubes, and — more than Math Nation credits — Unit 1 Lesson 2, where students turned repeating decimals into fractions. That was the proof that repeating decimals are rational, and it is what makes today's definition usable. Math Nation lists “N/A.”

Working toward MA.8.NSO.1.2 tomorrow, and MA.8.NSO.1.7 — order of operations with radicals — later in the course.

Where this sits in the course MA.8.NSO.1.1 is reported on the Grade 8 FAST under **Number Sense and Operations and Probability**, 22–28% of the exam. The estimating taught today is also what makes MA.8.NSO.1.2 possible, so between them these three lessons carry two benchmarks.

Materials Whiteboards and markers. Calculators — needed for the squaring and cubing in step 4, and deliberately not used for the bounding steps. Worksheet 7A.02.04. The perfect-square and perfect-cube anchor charts from Lessons 1 and 3.

Lesson at a Glance

Windy Hill periods run **53 minutes**.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up: same and different	2	4 min
Notes I–IV: rational decimals, irrational decimals, the radicand decides, four steps	3–6	13 min
Example 1: $\sqrt{13} \approx 3.6$	7–9	5 min
Example 2: $\sqrt[3]{20} \approx 2.7$	10–11	4 min
Example 3: Priya and her calculator — round last	12–14	5 min
Whiteboards: 9 questions, each with its reveal	15–32	16 min
IXL — start today's skills	33	5 min
Total		53 min

Where the lesson lives: Notes III (slide 5) and Example 3 (slides 12–14). Notes III is the sorting rule the FAST asks directly. Example 3 is the arithmetic trap that the textbook itself falls into.

Read this before you plan the day. Math Nation's component times total 45 to 60 minutes before any worksheet — at the upper bound, seven minutes longer than a full period with nothing assigned. The plan above fits 53 because their Collaboration and both Explorations are compressed into Notes II and III with Examples 1 and 2 as the worked versions.

On calculators. Math Nation's materials note says calculators should not be used “for the remainder of this lesson or for the subsequent lessons.” The instinct is right — estimating is the skill — but a three-lesson ban cuts against your standing policy, and as of 2 September nothing in these materials prints a calculator rule at all: calculators are allowed on everything, you announce it, and no page prints a rule about them. Worksheet and Additional Practice question 11 each mention a calculator DISPLAY, which is the scenario the question is about rather than an instruction about use. The estimating is protected by the QUESTIONS instead. Worksheet questions 4 and 5 ask for the perfect squares or cubes by name in part a before the root in part b, and the unit assessment's question 2 asks for the comparison alongside the bracket. A calculator does that arithmetic and still does not choose the comparison, which is the thing being taught.

No partner or group work anywhere in this lesson. Math Nation's version has seven printed instances plus a Think-Pair-Share. The 2.4.2 Collaboration is now Notes II and the whiteboard sequence; the 2.4.4 Exploration, which ends with “compare your answers to your partner's,” is now Notes III with Examples 1 and 2.

Vocabulary

rational number — a number that can be written as a ratio of two integers

irrational number — a number that cannot be written as a ratio of two integers

principal square root — the non-negative square root — the value the radical symbol returns

approximate — a value close enough to be useful, but not exact

A note on the glossary. Math Nation lists two key terms here, “irrational number” and “principal square root.” Both are above. “Principal square root” was already taught in Lesson 2 of these materials, so today it is a name being attached rather than an idea being introduced.

Slide-by-Slide Teaching Notes

Slide 1. Approximating Irrational Numbers 1 min

- Post the learning target.
- Call back to Lesson 1 explicitly — the side of a square with area 8 was between 2 and 3, and you told them it was not a whole number, not a fraction and not a decimal that stops.
- Today that number gets its name, and whiteboard question 6 gives it its value.
- New benchmark: MA.8.NSO.1.1, in the Number Sense reporting category, which is 22–28% of the exam.

Slide 2. Warm-Up 4 min

- TYPE: Same and different.
- INDEPENDENT.
- Expect: same — both are square roots, both radicands are in the forties, both values are near 7.
- Different — 49 is on the perfect-square list and 48 is not, so one is exact and one can only be approximated.
- The observation to fish for is that the two values are very close (7 and about 6.93) but completely different KINDS of number.
- This replaces Math Nation's estimation problem, which had no connection to roots.
- OFF THE SLIDE, YOURS TO SAY: One is a whole number. The other is not, and it never will be — however many decimals you write.

Slide 3. Notes: I. A rational number's decimal stops or repeats 3 min

- Students copy all three lines.
- The panel is the callback that makes this land: in Unit 1 Lesson 2 they turned $0.363636\dots$ into $\frac{4}{11}$ with algebra.
- That WAS a proof that repeating decimals are rational, and it is the reason item B is true rather than a rule they were handed.
- OFF THE SLIDE, YOURS TO SAY: Unit 1 turned every repeating decimal into a fraction. That is why they are rational.

Slide 4. Notes: II. An irrational number's decimal does neither 3 min

- THE NEW IDEA IN THIS LESSON.
- Item C is a named misconception on Florida's list — students think π equals 3.14 or $\frac{22}{7}$.
- Put both on the board next to $3.14159265\dots$ and ask which one is π .

- None of them.
- The panel is the honest version of the whole idea and it is worth saying slowly.
- OFF THE SLIDE, YOURS TO SAY: You can get as close as you like. You can never arrive.

Slide 5. Notes: III. The radicand decides, not the radical sign 4 min

- Students copy the table.
- This is the sorting rule and the FAST asks it directly.
- The one thing to say plainly: a radical sign is not a signal.
- Half the class will call every square root irrational unless you build this table with them.
- Point back at Lesson 1's list of perfect squares to 225 and Lesson 3's list of perfect cubes to 125 — those two lists ARE the answer key for this slide.
- OFF THE SLIDE, YOURS TO SAY: $\sqrt{\quad}$ of a perfect square, $\sqrt[3]{\quad}$ of a perfect cube. Check the number underneath.

Slide 6. Notes: IV. Approximating to the nearest tenth — four steps 3 min

- Students copy the outline.
- Insist on the order.
- The most common failure is jumping straight to testing decimals without bounding first, which turns a thirty-second problem into a two-minute one.
- Step 3 is optional in the sense that step 4 would find the answer anyway, but it halves the testing.
- Worksheet questions 4 to 7 are these four steps split into parts.
- OFF THE SLIDE, YOURS TO SAY: In this order, every time. Skipping step A is how students end up guessing.

Slide 7. Example 1 1 of the 5 min

- SPOKEN ONLY for the last line.
- Steps 1 and 2 are mechanical once the perfect-square list is known.
- The question at the bottom is step 3 and they should answer it before it is written down: $13 - 9 = 4$ and $16 - 13 = 3$, so it is nearer 4.

Slide 8. Example 1 2 of the 5 min

- Step 4 is where students want a calculator — give it to them for the squaring, and say so.
- The interesting moment is that 3.6 misses by 0.04 while 3.7 misses by 0.69, a seventeen-fold difference: the tenth is not a close call even though the whole numbers were.
- OFF THE SLIDE, YOURS TO SAY: Step 3 is why you start testing at 3.6 and not at 3.1.

Slide 9. Example 1 2 of the 5 min

- Plot it on the board line yourself as they watch.
- The point of the picture is that the answer is a POSITION, not just a decimal — which is what Lesson 5 will ask them to produce for seven numbers at once.
- OFF THE SLIDE, YOURS TO SAY: Between 3 and 4, nearer 4 than 3 — but only just past the 3.5 mark.

Slide 10. Example 2 *1 of the 4 min*

- SPOKEN ONLY.
- The error to head off is using 16 and 25 here because those are the squares nearest 20 — that is Sara's mistake in Lesson 6 and it is the most common error on this benchmark.
- Read the index first, every single time.
- OFF THE SLIDE, YOURS TO SAY: The index is 3, so the list is the perfect CUBES. That is the only thing that changes about the method.

Slide 11. Example 2 *3 of the 4 min*

- The only thing that changed is the list.
- Say that out loud and write both lists side by side on the board if they are not up already. 19.683 misses by 0.317 and 21.952 misses by 1.952, so 2.7 wins comfortably — but note that step 3 said 'nearer 3' and the answer is 2.7, which is nearer 3 than 2.
- Step 3 narrows where to look; it does not name the digit.

Slide 12. Example 3 *2 of the 5 min*

- Take a vote.
- Do not resolve it.
- Math Nation's own answer key for this lesson approximates $2\sqrt{6}$ as 4.8, and their own number line on the same page plots it at 4.9 — both cannot be right, and the textbook being wrong here is worth saying out loud.

Slide 13. Example 3 *2 of the 5 min*

- Do the subtraction in front of them: $2.449489 - 2.4 = 0.049489$, and doubling it gives 0.098978.
- That is what moved the answer from 4.9 to 4.8.
- The error is not carelessness; it is rounding at the wrong moment.

Slide 14. Example 3 *1 of the 5 min*

- The extra place is the practical rule, because students will sometimes have to round early.
- One more digit was enough here and it usually is.
- Worksheet questions 8 and 11 are both this idea, and the assessment asks it once.
- OFF THE SLIDE, YOURS TO SAY: Adding and subtracting are safe. Multiplying and dividing are not — round LAST, and if you must round early, carry one extra decimal place.

Slide 15. Whiteboards • Question 1 of 9 — ask *about 25 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- A board with only one of the two answers on it is incomplete, not wrong — say which before you start.

Slide 16. Whiteboards • Question 1 of 9 — reveal *about 1 minute with the discussion*

- The baseline.
- Exact, because 49 is on the list.
- Everything after this changes exactly one thing.

- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 17. Whiteboards · Question 2 of 9 — ask *about 25 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- A board with only one of the two answers on it is incomplete, not wrong — say which before you start.

Slide 18. Whiteboards · Question 2 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the radicand, by one. 48 is not on the list, so the answer stops being exact — 6.928..., which is 6.9.
- The board to look for says 6.9 with no mention of irrational, or 'irrational' with no value; the question asks for both.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 19. Whiteboards · Question 3 of 9 — ask *about 25 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- A board with only one of the two answers on it is incomplete, not wrong — say which before you start.

Slide 20. Whiteboards · Question 3 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the index.
- Same 48, different list, and the answer is roughly half.
- Anyone who uses 36 and 49 here has not read the index — that is Example 2's warning arriving on cue.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 21. Whiteboards · Question 4 of 9 — ask *about 25 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- A board with only one of the two answers on it is incomplete, not wrong — say which before you start.

Slide 22. Whiteboards · Question 4 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the radicand, to a number that IS on the cube list.
- Back to exact.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 23. Whiteboards · Question 5 of 9 — ask *about 25 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.

- A board with only one of the two answers on it is incomplete, not wrong — say which before you start.

Slide 24. Whiteboards · Question 5 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the index — and this time the answer stays rational.
- Apart from 1, 64 is the only number in this unit's range on BOTH lists — $1 = 1^2 = 1^3$ and $64 = 8^2 = 4^3$ — which is why 64 is here.
- A student who expects the index change to break exactness (as it did between questions 1 and 2) will hesitate, and hesitating is the right response to check.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 25. Whiteboards · Question 6 of 9 — ask *about 25 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- A board with only one of the two answers on it is incomplete, not wrong — say which before you start.

Slide 26. Whiteboards · Question 6 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT CLOSES THE UNIT'S OPENING.
- One feature changed: the radicand. 8 IS a perfect cube and is NOT a perfect square, so under a square root it is irrational — the trap in one question.
- And this is Lesson 1's square with area 8: you told them then that its side was not a whole number, not a fraction and not a decimal that stops.
- It is 2.828..., and now they can say what kind of number that is.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 27. Whiteboards · Question 7 of 9 — ask *about 25 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- A board with only one of the two answers on it is incomplete, not wrong — say which before you start.

Slide 28. Whiteboards · Question 7 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: it is not a radical at all.
- Every question so far has been a root, and a student can survive those by asking "is the radicand on the list" without ever meeting what irrational MEANS. π has no list to be on.
- THE BOARD TO LOOK FOR SAYS RATIONAL, BECAUSE $22/7$ — and that is worth two minutes, because $22/7$ is 3.142857..., a rational number close to π and not equal to it. π is not a fraction and cannot be written as one.
- The reference sheet says the Grade 8 FAST gives 3.14 or $22/7$ as APPROXIMATIONS to compute with; an approximation you are handed is not the number.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 29. Whiteboards · Question 8 of 9 — ask *about 25 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- A board with only one of the two answers on it is incomplete, not wrong — say which before you start.

Slide 30. Whiteboards · Question 8 of 9 — reveal *about 1 minute with the discussion*

- THE MIRROR OF QUESTION 7, and one feature changed: this decimal goes on forever TOO, and it is rational.
- A student who has decided that "never ends" means irrational will answer this one wrong having answered π right, which is why the pair sits together.
- They spent all of Lesson 1.02 turning this exact number into a fraction — $10x - x = 6$, so $9x = 6$ and $x = 2/3$ — and the fraction is the proof.
- THE BOARD TO LOOK FOR SAYS IRRATIONAL.
- Ask it to write 0.666... as a fraction; it can, and that settles it.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 31. Whiteboards · Question 9 of 9 — ask *about 25 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- A board with only one of the two answers on it is incomplete, not wrong — say which before you start.

Slide 32. Whiteboards · Question 9 of 9 — reveal *about 2 minutes with the closing discussion*

- WRITTEN WORK, and the only question in the round that asks for it.
- With the worksheet gone this is the only place in the period the habit gets built, and IXL takes an answer without ever seeing the work.
- ANSWER: no. 0.666... goes on forever and is $2/3$, which is rational.
- Going on forever is not the test; going on forever WITHOUT REPEATING is.
- THE MARK IS FOR THE COUNTEREXAMPLE, and the question says so — "no" on its own is a coin flip, and a named number that breaks his rule is an argument.
- Any repeating decimal from Lesson 1.02 works, and the students who reach for 0.333... or 0.999... are reaching correctly.
- If someone offers $0.999... = 1$, take it: a decimal that goes on forever and equals a whole number is the strongest version of this.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 33. IXL *5 min*

- IXL.
- The skills are assigned in Google Classroom, not named here, and they are deliberately not ours -- Croix: "The students need questions that aren't exclusively mine."
- Makes them think." Circulate.

- The thing to watch for is a student answering in the box with no paper in front of them: IXL takes an answer and never sees the work, so this is the one part of the period where the habit can quietly lapse.

Worksheet 7A.02.04 — Answer Key

1. four values	$\sqrt{16} = 4$ $\sqrt{-16} = \text{no real value}$ $\sqrt[3]{27} = 3$ $\sqrt[3]{-27} = -3$ Only the square root of a negative is refused.
2. which one, and why	$\sqrt{-16}$. A square root of a negative is never real. The point needs the reason. Any real number times itself is positive or zero.
3. select all	$\sqrt{18}$, $\sqrt[3]{30}$ and π Three of six. $\sqrt{25} = 5$ and $\sqrt[3]{27} = 3$ are whole numbers; $4/9$ is already a ratio.
4. $\sqrt{23}$	between 16 and 25; between 4 and 5; closer to 5 $23 - 16 = 7$ and $25 - 23 = 2$.
5. $\sqrt[3]{55}$	between 27 and 64; between 3 and 4; closer to 4 $55 - 27 = 28$ and $64 - 55 = 9$.
6. $\sqrt{78}$	$8.8^2 = 77.44$ (off by 0.56) $8.9^2 = 79.21$ (off by 1.21) $\sqrt{78} \approx 8.8$
7. $\sqrt[3]{120}$	$4.9^3 = 117.649$ (off by 2.351) $5.0^3 = 125$ (off by 5) $\sqrt[3]{120} \approx 4.9$ Let them use the calculator for the cubing and say so.
8. using 8.83	$\sqrt{78} \div 4 \approx 2.2$ $\sqrt{78} + 5 \approx 13.8$ $-3\sqrt{78} \approx -26.5$ The third one is why the sheet gives 8.83 rather than 8.8. Multiplying the rounded 8.8 by -3 gives -26.4 , off by a tenth.
9. A–D	A $\sqrt[3]{120} \approx 4.9$ B $\sqrt[3]{120} - 2 \approx 2.9$ C $2\pi \approx 6.3$ D $1 + \pi \approx 4.1$
10. the plot	B at 2.9, D at 4.1, A at 4.9, C at 6.3 The tick points are tenths, so all four land exactly on a mark.
11. Priya	The calculator is right. $2\sqrt{6} = 4.898\dots$, which is 4.9. She rounded first and then multiplied. Using $\sqrt{6} \approx 2.45$ gives 4.90. The point needs the diagnosis, not just the right number.

Question 11 is the one to read tonight. It asks students to diagnose an error, not just to compute. A student who writes “4.9” and stops has half of it; the other half is naming what Priya did wrong.

Question 3 is one of the three to circulate for; the others are 8 and 11. Select-all, and the two traps are $\sqrt{25}$ and $\sqrt[3]{27}$ — both whole numbers wearing a radical.

Additional Practice 7A.02.04 — Answer Key

1. four values	$\sqrt{81} = 9$ $\sqrt{-81} = \text{no real value}$ $\sqrt[3]{64} = 4$ $\sqrt[3]{-64} = -4$
2. why one and not the other	<i>Squaring gives a positive or zero; cubing a negative gives a negative.</i> Both halves are needed for the point.
3. select all	<i>$\sqrt{40}$, $\sqrt[3]{50}$ and 2π</i> $\sqrt{36} = 6$ and $\sqrt[3]{125} = 5$ are whole numbers; $3/4$ is a ratio. Scaling π by 2 leaves it irrational.
4. $\sqrt{95}$	<i>between 81 and 100; between 9 and 10; closer to 10</i> $95 - 81 = 14$ and $100 - 95 = 5$.
5. $\sqrt[3]{110}$	<i>between 64 and 125; between 4 and 5; closer to 5</i> Both cubes are on the Lesson 3 list. Math Nation's version used 141, which needs 216.
6. $\sqrt{95}$	<i>$9.7^2 = 94.09$ (off by 0.91) $9.8^2 = 96.04$ (off by 1.04) $\sqrt{95} \approx 9.7$</i> A close one — worth pointing out, since the whole numbers were not close at all.
7. $\sqrt[3]{110}$	<i>$4.7^3 = 103.823$ (off by 6.177) $4.8^3 = 110.592$ (off by 0.592) $\sqrt[3]{110} \approx 4.8$</i>
8. using 9.747	<i>$\sqrt{95} + 3 \approx 12.7$ $\sqrt{95} \div 2 \approx 4.9$ $2\sqrt{95} \approx 19.5$</i> Three decimal places on purpose. Rounding to 9.7 and doubling gives 19.4, a tenth short.
9. A–D	<i>A $\sqrt[3]{110} \approx 4.8$ B $\pi + 2 \approx 5.1$ C $3\pi \approx 9.4$ D $\sqrt{95} \approx 9.7$</i>
10. the plot	<i>A at 4.8, B at 5.1, C at 9.4, D at 9.7</i>
11. Marcus	<i>The calculator is right. $3\sqrt{7} = 7.937...$, which is 7.9.</i> He rounded first, then tripled, losing 0.137.

This sheet is optional. Post it to Google Classroom. It is not assigned and it is not collected.

Changes from the Math Nation Lesson

- **An arithmetic error was corrected.** Their 2.4.6 keys $2 \cdot \sqrt{6}$ at 4.8. The value is 4.89898..., which is 4.9 to the nearest tenth — and their own number line on the same page plots the point at about 4.9, so the table and the picture disagree. This is now slides 12–14 and worksheet question 11, taught as the rule “round last, not first.”
- **Two questions on the practice sheet asked the wrong thing.** Their questions 4 and 6 read “between which two perfect squares does $\sqrt{95}$ lie?” and “between which two perfect cubes does $\sqrt[3]{141}$ lie?” $\sqrt{95}$ lies between 9 and 10, and the question should have asked about the radicand: 95 is not a perfect square, which is what makes the root irrational. The homework sheet asks both correctly; the practice sheet did not. Both now ask about the radicand.
- **Two symbol slips in teacher notes were fixed.** Their 2.4.4 scaffolding asks “which cube root $\sqrt[3]{9}$ is closest to” when it means $\sqrt[3]{9}$, and their 2.6.2 key says “ $\sqrt[3]{49}$ is approximately -3.7 ” when it means $-\sqrt[3]{49}$.
- **The cube values were brought inside the taught list.** Their practice sheet uses $\sqrt[3]{141}$, which needs 216, and their 2.4.4 number line uses $\sqrt[3]{140}$, same problem. Both are retuned to $\sqrt[3]{110}$, which sits between 64 and 125 — the cubes students actually learned. Two larger cubes are kept, marked STRETCH, on the Lesson 6 circuit.
- **The three labels $\sqrt[3]{8}$, $\sqrt[3]{9}$ and $\sqrt[3]{10}$ were dropped from one number line.** In their 2.4.4 all three fall within 0.16 of each other on a line running 0 to 6, which is not plottable at that scale.
- Every partner and group activity is replaced — seven printed instances plus the Think-Pair-Share. The Collaboration and both Explorations are now Notes II and III with Examples 1 and 2.
- The warm-up was replaced. Math Nation’s number pyramid is a sound arithmetic puzzle with no connection to the lesson. The new one asks for $\sqrt{49}$ beside $\sqrt{48}$ — one radicand on the perfect-square list and one not — which is the sorting habit the lesson depends on.
- The wrap-up was replaced. Theirs asks students to “explain the process for approximating an irrational number,” keyed “answers will vary.” The Ticket Out the Door asks two sorting questions and one approximation, and it tells you who can bound before testing.
- The blanket calculator ban was not carried over. Math Nation bans calculators for this lesson and the two after it; nothing here prints a calculator rule at all, and the estimating is protected by the questions instead. See “On calculators” above.
- A select-all item was added, in the format the FAST uses.
- **Nothing else in the mathematics was changed.** Every value on their sheets is either still here or retuned for a reason named above.